

CONTACT INFORMATION	Department of Applied Mathematics The Hong Kong Polytechnic University Hung Hom, Kowloon, Hong Kong	Phone: (+852) 5309-0564 Email: keyu1.zhang@polyu.edu.hk Homepage: keyu-zhang523.github.io
RESEARCH INTERESTS	Stochastic Control and Optimization, Mean Field Games, Mathematical Finance, Reinforcement Learning.	
ACADEMIC APPOINTMENTS	The Hong Kong Polytechnic University - Postdoctoral Fellow Supervisor: Xiang Yu - Research Assistant Supervisor: Xiang Yu	Kowloon, Hong Kong July 2026–present July 2024–July 2025
EDUCATION	Tsinghua University - Ph.D. in Statistics, with Distinction - Advisor: Prof. Zongxia Liang Shandong University - B.S. in Mathematics and Applied Mathematics	Sep 2021–June 2026 Sep 2017–June 2021
PREPRINTS	[1] Erhan Bayraktar, Zhenhua Wang, Xiang Yu and Keyu Zhang (2026). Equilibrium for time-inconsistent mean field games: A systematic analysis by entropy regularization. Submitted. (arXiv:2605.14363) [2] Yu-Jui Huang, Xiang Yu and Keyu Zhang (2026). Policy iteration achieves regularized equilibrium under time inconsistency. Submitted. (arXiv:2603.06145) [3] Zongxia Liang, Xiang Yu and Keyu Zhang (2025). Mean field game with reflected jump diffusion dynamics: A linear programming approach. Major Revision with Mathematics of Operations Research. (arXiv:2508.20388)	
PUBLICATIONS	[4] Zongxia Liang, Xiang Yu and Keyu Zhang. On time-inconsistent extended mean-field control problems with common noise. Mathematics of Operations Research, online first, 2026. [5] Xiang Yu, Jiacheng Zhang, Keyu Zhang and Zhou Zhou. Major-Minor mean field game of stopping: an entropy regularization approach. SIAM Journal on Control and Optimization, 64(5): 3185–3214, 2026. [6] Zongxia Liang, Jianming Xia and Keyu Zhang. Equilibrium stochastic control with implicitly defined objective functions. SCIENTIA SINICA Mathematica, online first, 2026. [7] Zongxia Liang and Keyu Zhang. Time-inconsistent mean field and n-agent games under relative performance criteria. SIAM Journal on Financial Mathematics, 15(4): 1047–1082, 2024. [8] Zongxia Liang and Keyu Zhang. A mean field game approach to relative investment-consumption games with habit formation. Mathematics and Financial Economics, 18: 577–622, 2024.	

TEACHING
EXPERIENCE

- Teaching Assistant, Tsinghua University 2021–2024
- Courses: Calculus A1 (Fall 2021, 2022, 2023) and Calculus A2 (Spring 2022, 2023, 2024).

AWARDS

- Outstanding Doctoral Thesis (top 10%). Tsinghua University, 2026.
- Outstanding Graduates (top 4%). Tsinghua University, 2026.
- THU Comprehensive Excellence Scholarship, First Prize. Tsinghua University, 2025.
- THU Comprehensive Excellence Scholarship, First Prize. Tsinghua University, 2024.
- THU Comprehensive Excellence Scholarship, First Prize. Tsinghua University, 2023.
- THU Comprehensive Excellence Scholarship, Second Prize. Tsinghua University, 2022.
- THU Excellent TA Awards. Tsinghua University, 2021 and 2022.